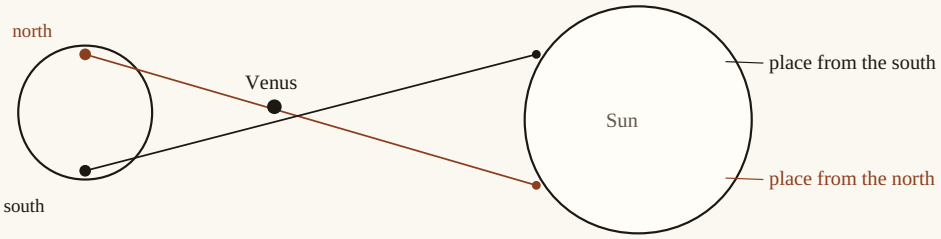
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

The Venus transits: a worldwide ruler

Kepler's clock gives the planets in units of Earth's orbit. A transit of Venus can turn that ratio into a length, if two stations far apart time the same event. What they compare is a differential parallax: Venus, being nearer, shifts on the Sun more than the Sun itself shifts. The full story of the ladder, including why 23 arcseconds is a one-radius illustration and not a ceiling, is the Venus-to-Cepheid lesson. This page is the human campaign.

What the stations compare

Schematic. Angles exaggerated. Not a traced 1769 track.



Differential parallax is about (1/0.277 - 1) times the solar parallax, about 2.61 times it.

One Earth radius, side-on, is about 23 arcseconds if the solar parallax is the modern 8.79.

That length is an illustration, not a ceiling. Stations can stand up to about a diameter apart.

Halley compared full internal-contact durations. One contact, timed in absolute time, needs longitude.

The black drop is mostly blur and the Sun's dim limb, not Venus's air by itself.

Two stations and the differential shift of Venus on the Sun

Teaching geometry, not a 1769 tracing. Halley compared durations of the internal contacts. A single contact timed on an absolute clock needs the longitude. The black drop is mostly blur and limb darkening.

Jeremiah Horrocks and William Crabtree, observing separately, saw the transit of 1639. Horrocks argued from Venus's small apparent disk, and from a model of sizes, that the solar parallax was not above 14 arcseconds, and the Sun at least about 15,000 Earth radii. That is not a two-station parallax. The book came out after his death, in 1662. Converted with modern constants, 14 arcseconds is about 58.4 million miles and 15,000 Earth radii about 59.4 million miles. Those mile figures are our conversion, not his words.

In 1761 Mason and Dixon were sent toward Bencoolen. Their ship, the Seahorse, was attacked in January. Mason's letter counts eleven killed and thirty-seven wounded. After repairs they observed from the Cape, and only the end of the transit, after sunrise on 6 June. Cloud early, then a clearing for the contacts. Do not promote that into a cloudless day. James Short compared the Cape with fifteen European stations. He was not their exclusive partner. His 8.47 and 8.52 arcseconds are for the day of the transit, not the mean distance. The mean-distance figure near 8.65 converts to about

94.5 million miles, roughly 1.7 percent high. A later Short value, 8.69, converts to about 94.1 million miles. Again, the miles are a modern conversion.

There was no one official answer. Pingre's 1765 reduction from Rodrigues, 10.25 arcseconds, converts to about 79.8 million miles, about 14 percent low. Hornsby, using 1769 and publishing in 1771-72, got a mean near 8.78 arcseconds, about 93.1 million miles, roughly 0.16 percent above today's 92.956 million miles. The astronomical unit is now defined as exactly 149,597,870,700 metres. Encke in 1824 combined both transits and got 8.5776 arcseconds, about 95.3 million miles, 2.5 percent high. That combination is not the 1769 result alone.

Some observers got little or nothing. Le Gentil, kept from Pondicherry by war in 1761, watched from a moving ship and the record was unusable. In 1769 a cloud hid Pondicherry again. Maskelyne at St Helena could not send a complete observation. Pingre at Rodrigues saw the Sun rise in thick cloud. At Greenwich, Bliss and Green missed the start for cloud.

In 1769 Cook and Green had a clear 3 June at Tahiti, after the quadrant was stolen on 2 May and recovered damaged but repairable. Green died of dysentery on the way home, on 29 January 1771. Wales and Dymond wintered at Hudson Bay and did observe. Their story is not a stopped clock. Mason's share of the Cape work is the next lesson. Dixon's sky in Norway, and whether anyone "won" against Chappe, is Dixon's lesson. This page will not rank them.

Chappe in Baja California, 1769

Pedro Espinoza Meléndez, writing in Meyibo, follows Chappe to San Jose del Cabo. The copy used here does not show the article's year or its pages. It says the party reached California on 19 May 1769, two weeks before the transit, after a month crossing the gulf. Chappe's diary, quoted there, has them camped on the shore and determined to reach San Jose the next day. Nineteen May is the landing the article dates. The walk into the town may be the day after. The article does not decide.

He set up in a granary, half the roof removed on the south side. Before 3 June he set the pendulum clock, checked the telescopes, and made repeated observations of the Sun. The transit was 3 June. On 5 June the Spaniards with the party fell ill, and so did the people of San Jose. Chappe nursed the sick and then became ill himself. The article does not date each of those days of care. On 18 June, fifteen days after the transit, he observed a lunar eclipse while very weak. Cassini, printing the register in 1772, says Chappe was then too weak to take the Moon at the meridian. That night was not the last line in the book. The same tables still enter later emergences of Jupiter's moons as Chappe's, including 20 June and 16 July. He did not leave Baja. He died on 1 August 1769. Counted on the calendar, that is fifty-nine days after the transit and seventy-four days after the 19 May landing. Those counts are arithmetic, not a sentence in either source. No date was found here for

when the survivors sailed. The Meyibo article says it is not clear whether the disease was typhus or yellow fever. This page says epidemic.

The article quotes Chappe's own instruments: a quadrant of three-foot radius by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. Dixon's weather, and the comparison of the two kits, is his lesson.

What the 18 June eclipse did not settle

The eclipse was a check on the longitude of San Jose. It did not become the longitude the book adopts, and it did not leave a measured improvement.

In *Voyage en Californie* (Paris, 1772) Cassini works the place in the astronomical observations, pages 88 to 92. Five emergences of Jupiter's moons, which he attributes to Chappe, average 7 hours 28 minutes 53 seconds west of Paris. The five include 20 June and 16 July, after the eclipse, so the average is not a baseline from before it. Cassini gives more weight to the transit. From the transit he has 7 hours 28 minutes 6 seconds, and he adopts 7 hours 28 minutes 10 seconds, which is 112 degrees 2 minutes 30 seconds west of Paris. Turning those hours into degrees is the time-to-longitude lesson. The worked example there uses Greenwich. This one uses Paris.

The eclipse, set only against a time calculated for Paris, gives 7 hours 29 minutes 7 seconds at the start and 7 hours 29 minutes 40 seconds at the end. Cassini says he knew of no observation of that eclipse at another known place. The two results stand 57 seconds and 90 seconds of time away from the figure he adopted. In angle, at fifteen seconds of arc to one second of time, that is 14 minutes 15 seconds and 22 minutes 30 seconds of longitude. The conversion is arithmetic done here. It is a disagreement. It is not a measured error, and it is not a measured improvement. No distance in miles is taken from it.

His remark that Jupiter's moons, without enough matched observations, should not be trusted closer than about 20 or 30 seconds of time is a caution about the method. It is not a personal error for Chappe. Nothing on these pages shows the eclipse tightening the longitude. The figure he asks the reader to keep is the one from the transit. A later ESO note says he stayed in order to fix the longitude. Staying made the check possible. It did not write the adopted value.

The illustrated lesson is download the PDF.

Sources

- Andreas Verdun, "The Determination of the Solar Parallax from Transits of Venus in the 18th Century," Archives des Sciences 57 (2004): https://www.unige.ch/sphn/Publications/ArchivesSciences/AdS%202004-2015/AdS%202004%20Vol%2057%20Fasc%201/45-68%20Verdun.pdf
- Nicholas Kollerstrom, on Horrocks, in the IAU transit volume: https://www.dioi.org/kn/IAUVenus-Transit.pdf
- James Short, Philosophical Transactions, 1761: https://royalsocietypublishing.org/rstl/article-pdf/doi/10.1098/rstl.1761.0102/1458661/rstl.1761.0102.pdf
- ESO transit background on the expeditions: https://www.eso.org/public/outreach/eduoff/vt-2004/Background/Infol2/EIS-F3.html
- Whipple Museum, the stolen Tahiti quadrant: https://www.whipplemuseum.cam.ac.uk/explore-whipple-collections/astronomy-and-empire/stolen-quadrant
- Pedro Espinoza Meléndez, "Medir el sistema solar. El viaje de Jean-Baptiste Chappe d'Auteroche a la Antigua California y la observación del tránsito de Venus en 1769," Meyibo (UABC). Year and pages are not shown on the copy used here: http://meyibo.tij.uabc.mx/index.php/meyibo/en/article/download/118/110/195
- Jean-Baptiste Chappe d'Auteroche, *Voyage en Californie*, Cassini's reductions, Paris, 1772, pp. 88-92: https://archive.org/download/voyageencaliforn00chap/voyageencaliforn00chap_djvu.txt
- ESO, on Chappe remaining for the 18 June 1769 lunar eclipse: https://www.eso.org/public/outreach/eduoff/vt-2004/Background/Infol2/EIS-F3_pf.html

Key facts

- Horrocks in 1639 set an upper bound near 14 arcseconds from Venus's apparent size, not from a second station. Publication was posthumous, 1662.
- Mason and Dixon reached the Cape after the Seahorse fight (11 dead, 37 wounded) and timed the end of the 6 June 1761 transit.
- Short, Hornsby, Pingre, and Encke published different parallaxes. Converted miles on this page are modern arithmetic, not their sentences.
- The modern au is exactly 149,597,870,700 metres, about 92.956 million miles. No 1761 or 1769 paper printed that definition.

- Chappe's party reached the Baja shore on 19 May 1769. The diary leaves open whether the town was the next day. Transit on 3 June, lunar eclipse on 18 June while he was very weak, death there on 1 August, of an epidemic. Later Jupiter-moon timings are still entered as his.
- Cassini, *Voyage en Californie* (1772), pp. 88-92, adopts 7 hours 28 minutes 10 seconds west of Paris from the transit. The eclipse was a check against a calculated Paris time. No measured improvement is given.